

repetitive-programmable tasks. When pressed, it provides immediate physical feedback and triggers immediate action. A smartphone falls short when you compare the number of steps needed to get the action running and, of course, the ease of configuration and use. Designed with simplicity in mind, One Button is the creators' effort at bringing the IoT to mainstream users.

Overcoming the initial hurdles

T. Anand, founder and CEO of Knewron Technologies, and his team of developers, talk about how they have put together the button. "The first and foremost task was to decide on the dimensions and shape of 1btn; good enough for people to use it in their daily lives and house or offices easily, without compromising on the aesthetics of their own stuff. Cramping many features and capabilities in that size was a challenge, thereafter," Anand says.

"We learnt from competitors about issues their products have and reviews by existing users, and accordingly worked out our offerings. There were many power-saving battery-charging-related problems and, thus, it took us a while



Just press to trigger the button

to decide on the right approach in making it work better, using lesser power," he adds.

From pressing the button to seeing the LED glow

At the heart of the button is a Wi-Fi system on chip (SoC), ESP8266, which is a Wi-Fi module in itself and has an integrated Transmission Control Protocol/Internet Protocol (TCP/IP) stack that helps connect any microcontroller to the Wi-Fi network. Pre-programmed with attention command (AT command) set firmware, the SoC has powerful onboard processing and storage capabilities.

The very first usage of a new button demands configuration to connect to your home/office Wi-Fi. Register online, create your user account and add the device to your list. A unique MAC ID differentiates

each button and you can even name it for easy recall.

On being pressed, the button wakes up from sleep, connects to the Internet via the Wi-Fi router and sends a service request to the API. The request is then processed and a pass/fail is sent back to the button, which displays a green/red light, correspondingly. All this happens in less than seven seconds.

With a rechargeable lithium-polymer battery inside, the device can be easily charged. One charge certainly can go way beyond 350-plus presses and more, say the developers, which for a typical usage means about two months.

The way forward

Launched in January 2016 at Electronics Rocks and IoTShow. in conference, the makers are offering at minimal cost and one-year-free back-end services. There is an option of using multiple buttons to work together by integrating ReST API calls into the application and accordingly programming 1btn. One can even avail only back-end services to go with one's own 1btn via a subscription model.

The team is also working on integrating Global System for Mobile Communications and Bluetooth Low Energy into 1btn to advance its capabilities. Having come up with a beta version in just three months, 1btn is on its way to becoming a completely open source product. **EFY**

The developers speak

"While 1btn has a simplified interface, it packs a lot of punch, especially with its open source design and multitude of design/configuration options. This is especially for hackers and makers.

"You can completely rewrite 1btn firmware and make it do something else of your liking. Even better, you can modify API end-points in their native firmware and re-route these somewhere else.

"On the other hand, if you wish to use existing infrastructure along with your own, you could redirect button press action to your personal cloud URL and then invoke some intelligent script written by you to do the heavy-lifting. You can also modify or retrofit 1btn to be used as a home automation/monitoring device for certain applications. For example, button contacts from 1btn could be taken out in the form of probes and used as contact/no-contact sensors.

"Compared to other global competitor variants, 1btn is scalable and programmable. Unlike many other buttons that are pre-programmed and bound with certain APIs/interfaces all their lifetimes, 1btn comes in a fully-programmable condition. You really own the hardware!"